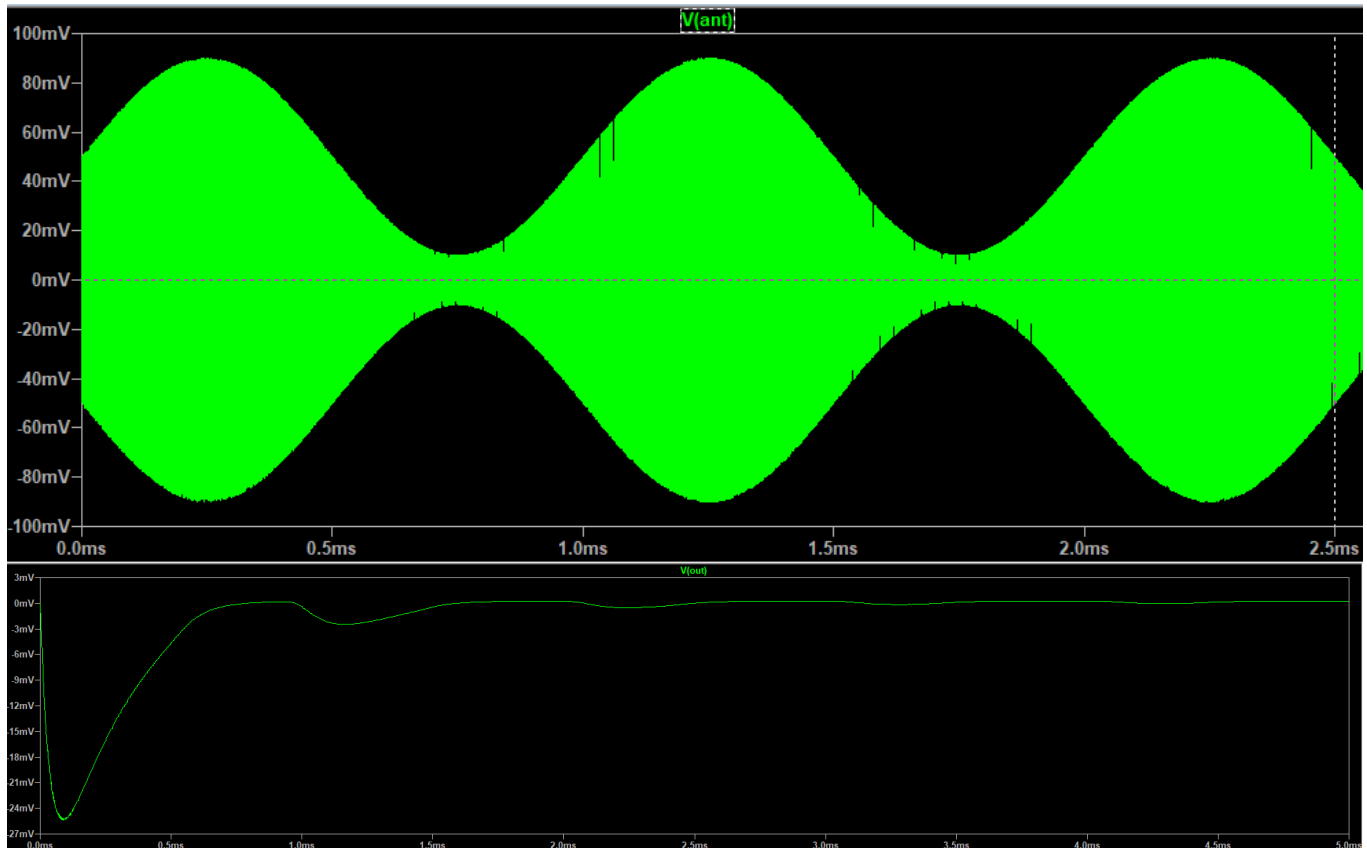


Ideal_LTSpice_Netlist_5

Prompt

The output doesn't seem to have a normal behaviour, could you explain why, fix the problem and make the appropriate modifications needed for the netlist



Answer

You have hit another excellent and very common hurdle in electronics design. Your analysis is spot on: the output behavior is not normal. The signal has completely flatlined into the low millivolt range.

The Diagnosis: Impedance Mismatch

The problem here is not your detector or your signal; the problem is the **32 Ω headphones** (R_LOAD).

Your audio amplifier is a simple "Common Emitter" pre-amplifier. The output resistance of this stage is largely determined by the 4.7 k Ω collector resistor (R_AUD_C). You are trying to drive a 32 Ω load with a 4.7 k Ω source.

When you connect standard $32\ \Omega$ headphones to this circuit, they act almost like an AC short-circuit to ground. The $32\ \Omega$ load demands far more current than this small transistor stage can physically provide. The voltage gets "pulled down" to practically zero, which is exactly why your output plot is a flatline.

This is why vintage simple AM radios historically used special "high-impedance crystal earpieces" (often $10\ \text{k}\Omega$ to $20\ \text{k}\Omega$) rather than modern low-impedance headphones.

The Fix

To see the beautiful 1 kHz audio sine wave in your simulation, we need to do two things:

1. **Change the Load:** We will change `R_LOAD` from $32\ \Omega$ to **$10\ \text{k}\Omega$** . This simulates the input of a modern powered speaker (like PC speakers), a dedicated audio amplifier IC, or a vintage high-impedance earphone.
2. **Add an Audio Emitter Bypass:** Just like we did for the RF stage, we will add a $10\ \mu\text{F}$ capacitor across the audio transistor's emitter resistor to unlock its maximum audio voltage gain.

The Modified Netlist

```
* AM Radio Receiver - Corrected for Impedance Mismatch
* -----
* POWER SUPPLY
V1 VCC 0 9V

* ANTENNA & AM SOURCE (Tuned to BBC 5 Live: 693 kHz, 1 kHz Tone)
B_ANT ANT 0 V=0.05 * (1 + 0.8*sin(2*pi*1k*time)) * sin(2*pi*693k*time)
R_ANT ANT TANK_IN 50

* LC TUNING STAGE (693 kHz)
L_TUNE TANK_IN 0 200u
C_TUNE TANK_IN 0 264p

* RF AMPLIFIER (NPN)
C_RF_IN TANK_IN RF_B 10n
Q_RF RF_C RF_B RF_E NPN_IDEAL
R_RF_B VCC RF_B 1Meg
R_RF_C VCC RF_C 4.7k
R_RF_E RF_E 0 1k
C_RF_BYPASS RF_E 0 100n

C_RF_OUT RF_C DET_IN 10n

* ENVELOPE DETECTOR (Demodulator)
```

```

D_DET DET_IN DET_OUT 1N34A
R_DET DET_OUT 0 10k
C_DET DET_OUT 0 10n

* AUDIO AMPLIFIER (NPN)
C_AUD_IN DET_OUT AUD_B 10u
Q_AUD AUD_C AUD_B AUD_E NPN_IDEAL
R_AUD_B VCC AUD_B 1Meg
R_AUD_C VCC AUD_C 4.7k
R_AUD_E AUD_E 0 1k
C_AUD_BYPASS AUD_E 0 10u      ; <--- MODIFICATION 1: Audio Gain Bypass
C_AUD_OUT AUD_C OUT 10u

* LOAD (Line-Out / High-Z Earphone)
R_LOAD OUT 0 10k              ; <--- MODIFICATION 2: Fixed Impedance Mismatch

* MODELS
.model NPN_IDEAL NPN(Bf=100)
.model 1N34A D(Is=200n Rs=2 N=1.2 Cjo=1p Eg=0.67)

* SIMULATION COMMAND
.tran 0 5m 0 0.1u
.end

```